Semantic Graph-Text Alignment for Interpretable and Drift Aware Network Intrusion Detection

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Introduction

Modern NIDS struggle with dynamic, real-world traffic where

- concept drift is constant
- labels are scarce.

Graph-based approaches (LAGER) capture evolving topologies and operate online, but rely only on statistical features, offering little semantic understanding.

In contrast, semantic encoders (PacketCLIP) align traffic with natural language descriptions, enabling interpretability, but lack online adaptability.

The challenge is to combine online graph dynamics with semantic embeddings to achieve adaptive and interpretable intrusion detection.

Problem Statement

- This work focuses on **online traffic processing**, using **graph-based representations enriched with semantic vectors**, and on **characterizing drift** as *benign or malicious*.

- The scope includes validation on **real-world datasets**, scalability analysis, and interpretability for analysts.

- The central problem is that current systems either **detect drift without meaning** (statistical models) or **provide meaning without adaptation** (semantic models). A solution must address both dimensions simultaneously.

Problem Background

Building an **online, semantically enriched NIDS** involves tackling several open challenges:

- **High-volume semantic encoding**: Network traffic is *massive and noisy*; generating **compact, multimodal embeddings** that preserve both protocol-level details and semantic meaning is computationally demanding.
- **Fusion of graph and semantic features**: Statistical graph dynamics capture *who communicates with whom*, while semantic embeddings capture *what the communication means*. Designing a **joint representation space** without redundancy or overfitting is non-trivial.
- **Robust drift characterization**: Real networks evolve continuously. The system must **differentiate benign evolution** (new devices, services) from **adversarial drift** (botnets, scans) while maintaining *plasticity* (adapt quickly) and *stability* (resist poisoning).
- **Scalability and efficiency**: **Real-time updates** of large dynamic graphs combined with semantic inference introduce significant overhead. Solutions require *graph sampling, embedding compression, and edge-optimized models*.
- **Interpretability and trust**: Analysts need **clear explanations**—not just *which device is anomalous* but also *why it is anomalous*, combining graph evidence with semantic labels for actionable insights.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
1	Ryozo Masukawa et al., 2025	PACKETCLIP	A multi-modal framework that combines packet data with natural language semantics through contrastive pretraining and hierarchical GNN reasoning. It improves traffic classification accuracy while providing human operators with natural language explanations of packet behavior, enhancing interpretability.	Traditional models struggle to interpret the semantics of network behaviors, and reasoning about the semantics of cyberattacks remains a key research challenge

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
2.	He et al., 2025	LAGER - Layer-wise Graph feature Extractor using Graph Neural Networks for online, unsupervised network intrusion detection with concept drift adaptability	Novel graph-based feature extraction framework that models networks as layer-wise attributed graphs (TCP, UDP, ARP, ICMP) and uses GNNs to generate representative embeddings. Introduces Concept Drift Compatibility Index (CDCI) for evaluating drift detection capability in NIDS. Addresses limitations of statistical feature extractors by providing neighborhood-aware, protocol-specific representations.	Lacks semantic understanding of detected patterns - only provides statistical anomaly scores without explaining what anomalies represent. Cannot distinguish between benign concept drift (new streaming service) and malicious activities (coordinated attacks) beyond statistical differences. Limited interpretability despite graph visualization capabilities.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
3	Zhenlu Sun, André M. H. Teixeira, Salman Toor, 2024	GNN-IDS (Graph Neural Network-based Intrusion Detection System)	Combines attack graphs with real-time measurements using GNNs (GCN, GAT). Captures both topological and temporal patterns, lowering false positives compared to traditional ML-based IDS.	Depends on predefined attack scenarios and manual attack graph construction. Does not adapt to novel attack patterns and lacks semantic interpretation beyond visualization.
4.	Susik Yoon, Youngjun Lee, Jae-Gil Lee, Byung Suk Lee, 2022	ARCUS (Adaptive Framework for Online Deep Anomaly Detection Under Complex Evolving Data Stream)	Employs adaptive pooling of multiple autoencoder-based models to manage evolving data streams. Handles multiple types of concept drift (sudden, gradual, incremental, recurrent) in IoT and cloud data.	(i) No semantic understanding, relies on anomaly scores. (ii) Reactive rather than proactive in handling drifts. (iii) Limited to autoencoder architectures, reducing general applicability.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
5	Ragini Gupta, Shinan Liu, Ruixiao Zhang, Xinyue Hu, Xiaoyang Wang, Hadjer Benkraouda, Pranav Kommaraju, Phuong Cao, Nick Feamster, Klara Nahrstedt, 2025	NetGuard - Generative Active Adaptation Framework for Drifting and Imbalanced Network Intrusion Detection	Employs density-aware active learning to identify informative samples for labeling and leverages deep generative models (VAEs, RealTabFormer) to synthesize diverse samples for data augmentation. Addresses concept drift, class imbalance, multi-class support, and labeling cost simultaneously. Boosts F1-score from 0.60 to 0.86, with rare attacks improving from near-zero detection (0.001-0.04) to 0.30-0.71 F1-scores	Focuses primarily on generative augmentation and active learning but lacks multi-modal cross-domain alignment between packet data and semantic descriptions. Does not integrate graph-based reasoning or natural language understanding for interpretable traffic analysis. Limited to traditional ML approaches without leveraging modern cross-modal representation learning techniques.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
6.	Zeyi Li, Minyao Liu, Pan Wang, Wangyu Su, Tianshui Chang, Xuejiao Chen, Xiaokang Zhou, 2025	Multi-ARCL (Multimodal Adaptive Relay-based Distributed Continual Learning) for Encrypted Traffic Classification	Implements distributed continual learning framework specifically designed for encrypted traffic classification. Addresses both Domain Incremental Learning (DIL) and Task Incremental Learning (TIL) scenarios. Uses multimodal adaptive relay mechanisms for knowledge transfer across tasks while maintaining privacy in distributed environments. Proposes adaptive relay-based method that eliminates silent neurons and enables effective model retraining.	Primarily focuses on continual learning mechanics rather than semantic interpretability. Lacks integration of natural language understanding or cross-modal alignment between network traffic and textual descriptions. Does not address real-time concept drift detection or provide semantic reasoning capabilities for traffic behavior understanding. Limited interpretability for human analysts in cybersecurity contexts.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
7	Jiangtao Zhai, Haoxiang Sun, Chengcheng Xu, Wenqian Sun, 2023	ODTC (Online Darknet Traffic Classification) based on Multimodal Self-Attention Chaotic Mapping Features	Integrates CNN and BiGRU for local space-time feature extraction from packet payload content, while using MLP for flow-level abstract features. Introduces feature amplification module using logistic chaotic mapping to enhance fuzzy features. Employs multi-head attention mechanism to discover hidden relationships between different features. Implements class incremental learning with weighted loss functions, achieving 50% reduction in time and memory consumption while improving classification accuracy	Focuses specifically on darknet traffic rather than general network traffic classification. Lacks semantic reasoning capabilities and natural language integration for interpretable analysis. Does not address concept drift adaptation or cross-modal alignment between packet behaviors and textual descriptions. Missing real-time drift detection mechanisms and human-interpretable explanations for traffic classification decisions.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
8.	Ippokratis Koukoulis, Ilias Syrigos, and Thanasis Korakis, 2025	Self-Supervised Transformer-based Contrastive Learning for Intrusion Detection Systems	A novel transformer-based framework that extracts flow representations directly from raw packet sequences using a self-supervised contrastive learning process. It is capable of detecting anomalies, improving performance with limited labeled data, and adapting to traffic from various domains.	• Random packet segment augmentation can be ineffective if the segments are too similar or too different from the original, leading to poor model learning and weak attack detection. • Simplified tokenization using one token per packet risks losing fine-grained byte-level information, potentially missing critical low-level attack signatures.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
9	Sadaf Sattar et al., 2025	ET-SSL (Encrypted Traffic Anomaly Detection using Self-supervised Contrastive Learning)	A novel self-supervised learning framework that uses contrastive learning to detect anomalies in encrypted network traffic without needing labeled datasets or payload analysis. It achieves high accuracy and a low false positive rate , can process traffic in real-time at high speeds , and detects zero-day attacks effectively.	It requires optimization for high-speed real-time deployment and that its study is based on simulated traffic rather than real-world encrypted traffic.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
10.	Alwhbi, I.A., Zou, C.C., Alharbi, R.N. (2024) [Sensors 24, 3509]	Survey of machine learning and deep learning techniques for encrypted network traffic analysis and classification; Focus on IoT, mobile, website traffic	Comprehensive review of machine-learning-based techniques for encrypted traffic classification, including data collection, feature extraction, model training; Analyzes applications in IoT, mobile, and web security contexts	The proposed models face several gaps, including challenges with dynamic networking environments (asymmetric routing, NAT, tunneling), high-dimensional data and computational overhead for real-time processing, limited frameworks and tools for large-scale deployment, vulnerability to obfuscation and evasion techniques, reduced accuracy from traffic sampling at high speeds, inadequate generalization to new types of encrypted traffic, and a lack of representative, labeled datasets for model training and evaluation.

Literature Review

S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
11.	X. Tan et al. (2024 / ACM proceedings)	Survey of contrastive learning approaches for network intrusion detection; Self-, semi-, and supervised contrastive learning	Proposes a generic framework and comprehensively reviews advances, applications, and architectures of contrastive learning for NIDS. Shows improvements in detection accuracy, robustness, and zero-day threat identification due to reduced reliance on labeled data	Identifies persistent challenges including data imbalance (more benign than malicious events), difficulty detecting infrequent attacks, challenges with highly similar benign/malicious behaviors, dependence on negative sample selection and surrogate task design, and high computation overhead for large-scale or real-time deployment
12.	Paul R. B. Houssel, Priyanka Singh, Siamak Layeghy, Marius Portmann, 2024 [arXiv]	Evaluates GPT-4 and LLama3 LLMs for network intrusion detection (NIDS); compares with classical and transformer models	First exploratory study showing LLMs are unsuitable as sole NIDS classifiers but promising as explainability agents; provides empirical results and discussion on computational cost, zero-shot, and fine-tuned scenarios	LLMs perform poorly as primary NIDS detectors on NetFlow data, struggle to detect domain-specific attacks, exhibit high inference latency, and suffer from “hallucination” and inconsistent explanations. Their practical use is limited to complementary roles, especially explainability, not real-time detection.

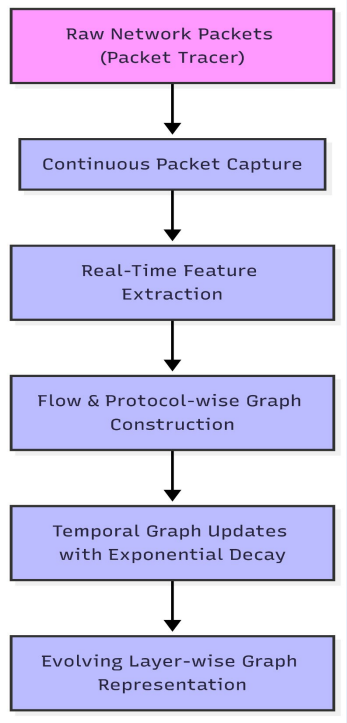
Literature Review

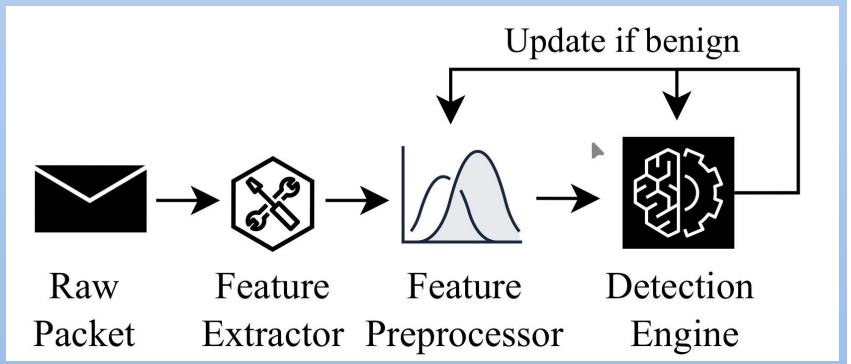
S. No.	Author(s), Year	Method/Approach	Key Contribution	Gaps Identified
13.	Misbah Shafi, Rakesh Kumar Jha, Sanjeev Jain, 2022	LGTBIDS (Layer-wise Graph Theory-Based Intrusion Detection System)	Models wireless communication networks as graphs, performs independent layer-wise analysis to identify attacked devices. Targets Beyond 5G networks with focus on spoofing and signal strength attacks while ensuring low computational complexity.	Relies only on energy efficiency and secrecy rate metrics. Lacks semantic understanding of attack patterns, uses simple statistical measures, and is constrained to wireless networks.

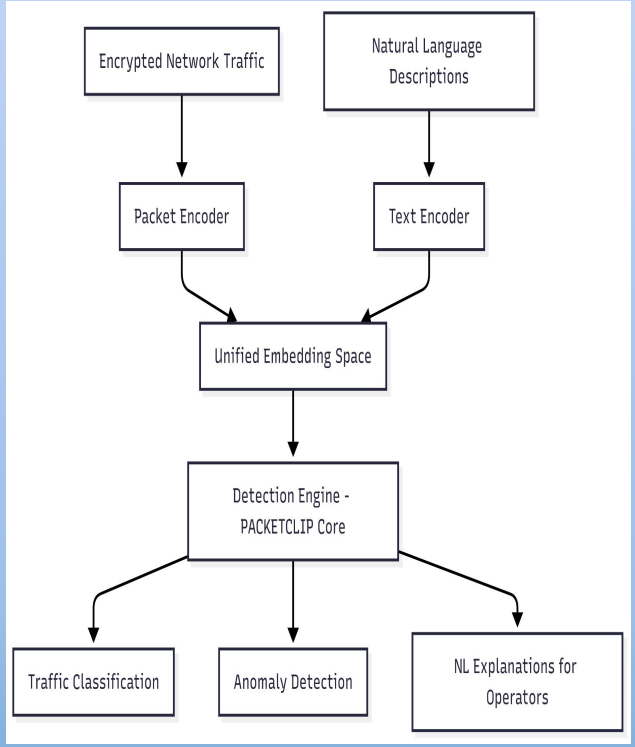
Existing Solution

- Graph Neural Networks for IDS - Strong anomaly detection but lack semantic interpretation beyond statistical anomaly scores
- Multi-modal Traffic Analysis - Aligns packet data with natural language but limited to packet-level classification, not topology reasoning
- Self-supervised Learning - Learn robust representations from unlabeled data but miss cross-modal graph-text alignment
- Drift Adaptation Frameworks - Handle evolving patterns reactively but cannot explain semantic causes of concept drift
- Post-hoc Explainability - Generate explanations after detection with high latency; not suitable for real-time integrated reasoning

Proposed Architecture

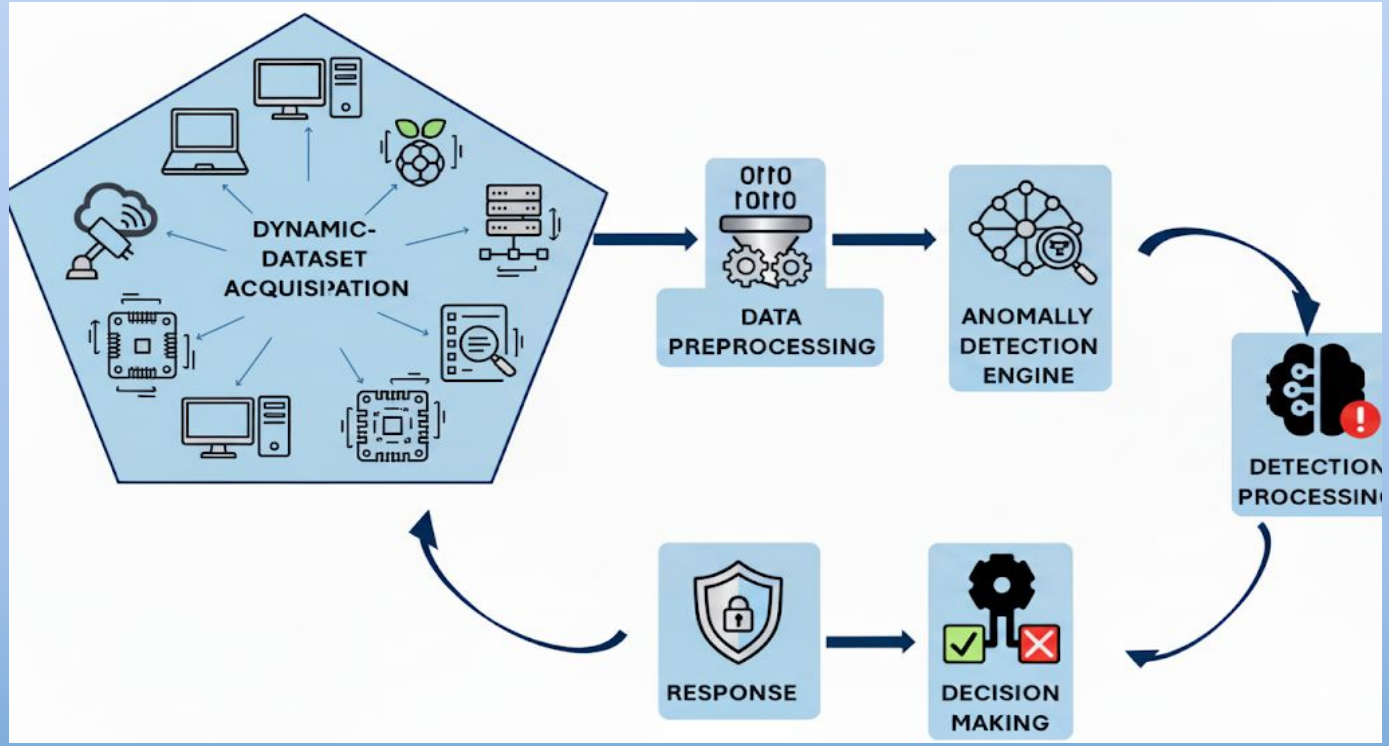




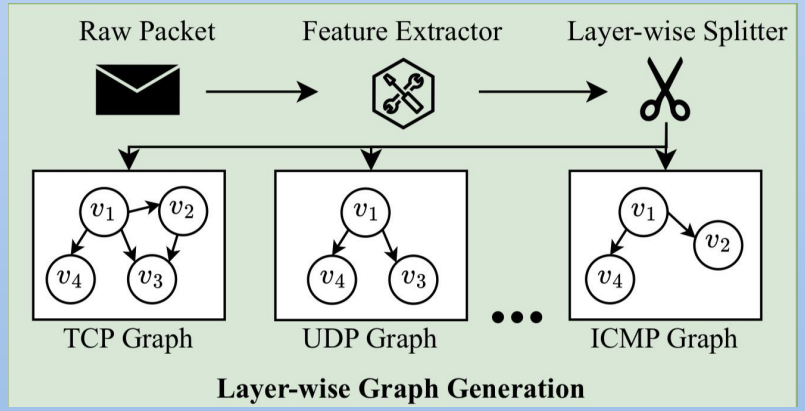


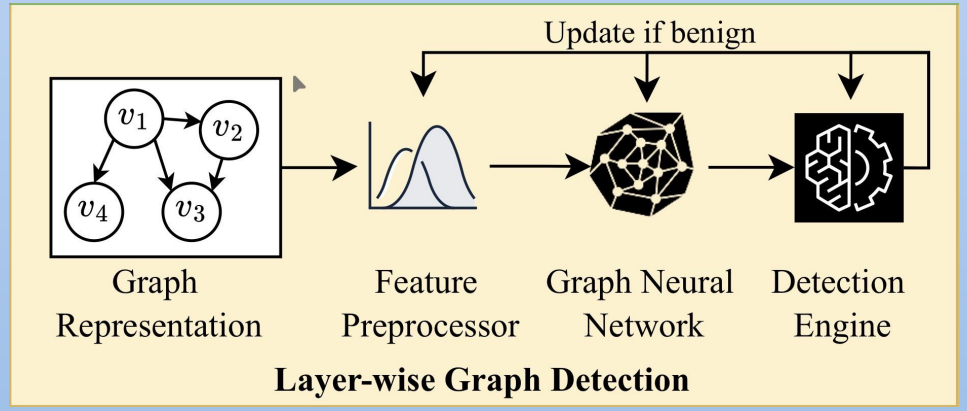
Proposed Architecture & Methodology

- Dataset Acquisition pipeline

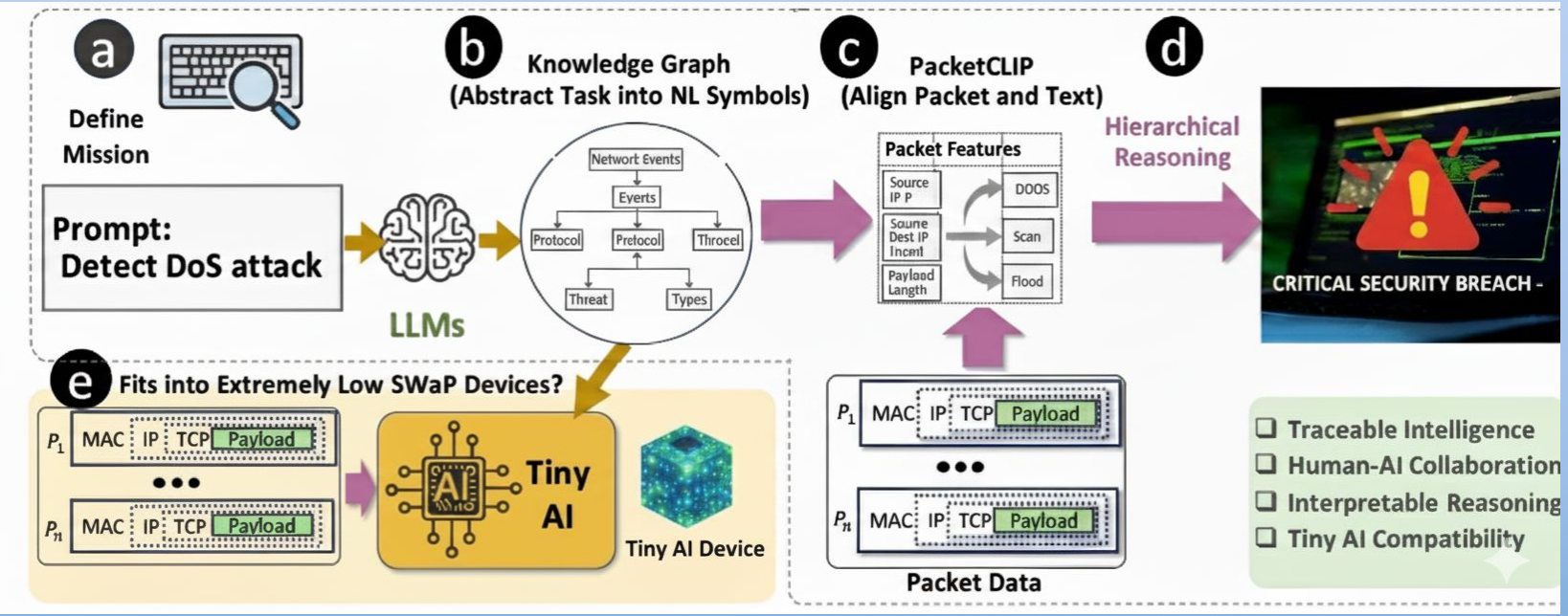


Proposed Architecture





Proposed Architecture & Methodology



Proposed Methodology

1. Real-World Dataset Utilization

- Modern intrusion detection requires real-world datasets, not static, synthetic collections.
- Real traffic captures:
  - Temporal dynamics of evolving threats
  - Irregularities of genuine user behavior
  - Concept drift as networks and attack strategies change
- Live data streams reflect authentic network topologies, volumes, and correlations, providing a realistic training ground and practical benchmark.
- Feasibility is achieved through:
  - Continuous packet capture
  - Real-time feature extraction
  - Ground truth labels maintained via analyst validation and external threat intelligence feeds

Proposed Methodology

2. Layer-Wise Graph Processing

- Traffic is transformed into graph representations, separating communication into protocol-specific layers (e.g., TCP, UDP, ICMP), which reduces complexity and highlights protocol patterns.
- Graphs are dynamically updated as new packets arrive, using incremental updates and exponential decay functions to emphasize recent behaviors while retaining historical context.
- This allows for constant-time updates of flow statistics, inter-arrival times, and structural properties like node centrality, providing an evolving view of network activity.
- Graph neural networks are used to extract meaningful representations.
- Message passing aggregates information from neighbors, and temporal aggregation captures feature evolution over time.
- Multi-layer processing combines local patterns (e.g., anomalous connections) with global trends (e.g., coordinated attacks).
- The result is a rich embedding space that encodes both structural and temporal aspects of the network.

Proposed Methodology

3. Cross-Attention with Semantic Integration

- Graph embeddings capture statistical and structural behaviors but lack inherent interpretability.
- A cross-attention mechanism integrates graph representations with semantic embeddings from language models to bridge this gap.
- The approach uses a dual-encoder design: one for traffic graphs, the other for natural language descriptions of attack and benign behaviors.
- Contrastive pre-training aligns these spaces, grouping similar behaviors near relevant textual descriptions.
- Cross-attention directly fuses these representations, with network embeddings as queries and semantic embeddings as keys and values.
- The enriched embeddings combine statistical evidence with contextual meaning.
- This integration supports zero-shot classification for unseen attack types by referencing textual descriptions.
- It also provides interpretability, distinguishing between superficially similar patterns (e.g., high packet rates from video streaming vs. DDoS attacks) by grounding them in semantic context.

Research Objectives

The primary goal is to create an adaptive, interpretable NIDS that unifies graph-based online learning with semantic embeddings. Specifically, the objectives are to:

- Develop a unified framework where evolving traffic graphs are enriched with contrastive semantic vectors, combining topology and meaning for richer detection signals.
- Design drift-aware mechanisms that not only detect anomalies but also characterize drifts in real time as benign behavior or malicious activity.
- Evaluate extensively on real-world datasets, benchmarking against purely statistical and purely semantic baselines to measure accuracy, robustness, and interpretability.
- Introduce continual learning capabilities, enabling the system to incrementally adapt to new benign patterns without retraining from scratch.
- Optimize scalability for deployment, using techniques like graph sampling, embedding compression, and edge-friendly inference to handle enterprise-scale traffic.

Proposed Timeline - 6 MONTHS OVERVIEW

S. No.	Duration	Focus	Milestone
1	Mid Jan 2026 – Feb 2026	Core Model Implementation	Implementation of the complete proposed system
2	Feb 2026 – Early Mar 2026	Model Training & Optimization	Model training, tuning, and stability validation
3	Early Mar 2026 – Mid Mar 2026	Evaluation & Analysis	Quantitative and qualitative evaluation, error analysis
4	Mid Mar – Early Apr 2026	Finalization	Documentation, report preparation, submission to a targeted Q1 journal, and final presentation

Tools and Technology

Programming Language: Python

Deep Learning Framework: PyTorch, PyTorch Geometric

Pretrained Models:

- GAT (Graph Attention Network – Graph Encoder), BERT-base-uncased (Text Encoder – Cross-Attention Learning), Flan-T5-Large (Text Augmentation)

NLP Libraries: Hugging Face Transformers, Tokenizers

Learning Techniques:

- Multi-Task Self-Supervised Learning (Attack Classification, Link Prediction, Community Detection, Structural Properties), Cross-Attention Mechanism (Bidirectional Graph-Text Alignment), Contrastive Learning (InfoNCE Loss)

Evaluation Metrics: Attack Classification F1-Score (Primary), Silhouette Score (Clustering), Davies-Bouldin Index, Link Prediction AUC, Cross-Attention Alignment

Experiment Tracking: TensorBoard

Hardware & Environment: GPU-based (CUDA/MPS Support)

Utilities: NetworkX, NumPy, Pandas, PyArrow, Scapy